Source-style function on the actual weighted-Proj curve

Proof packet | 28 October 2026 | pinned Lean 4.34.1

Checked construction

Under Field K and separability of the quintic, the prior whole-curve IsIntegral theorem supplies a genuine Scheme.functionField. The actual Z-basic open is nonempty. The new chart-to-function-field map is injective, so the affine x and y coordinates have defined images in this field.

The checked source numerator is N = B(s1*x+t1) - (C*x+A)(x^2+q1*x+r1), with B,A,C from Common22SupportHead.coefficientData. The checked field expression is phi(N)/phi(B)-phi(y). With the explicit hypothesis B != 0, the denominator image is nonzero.

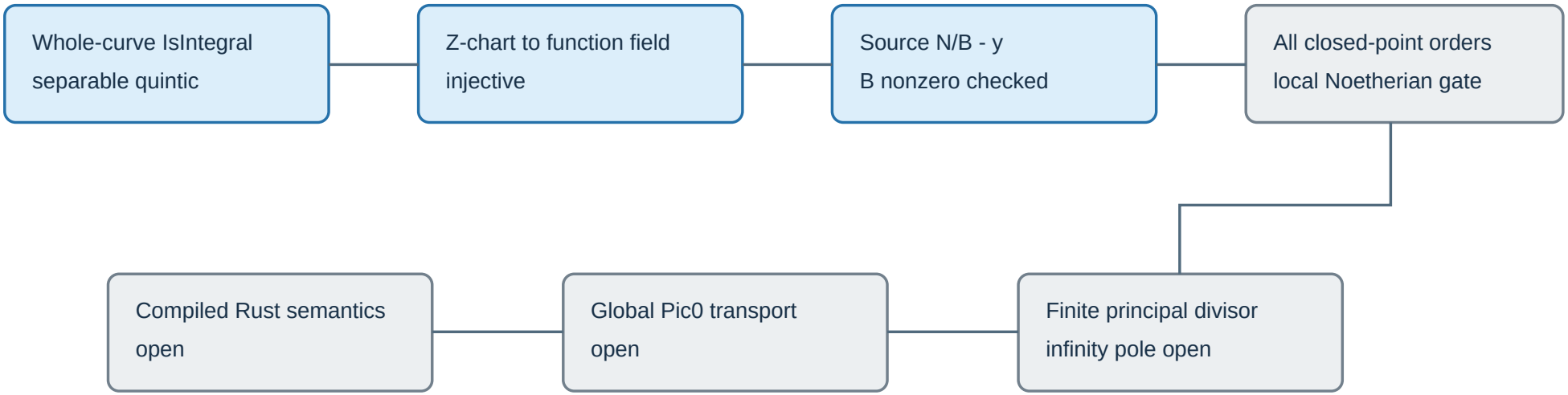
Exact boundary

The direct axiom print contains only propext, Classical.choice and Quot.sound. No source support-validity or global divisor-class claim follows from this definition. This packet contains no DLP experiment or cost comparison.

Next proof gates

Mathlib OrderOfVanishing requires IsLocallyNoetherian X; its non-junk order uses coheight z = 1. Neither local Noetherianity nor all-closed-point coheight, finite support, infinity pole, principal divisor, Pic0 class transport, or compiled Rust correspondence is established here. The CSV distinguishes checked and open items.

Actual weighted-Proj source function: checked boundary



Blue: kernel-checked declaration. Gray: open proof obligation. Arrows: dependency, not measured cost.

Blue: checked declaration. Gray: open gate. Arrows: proof dependency, not performance.